

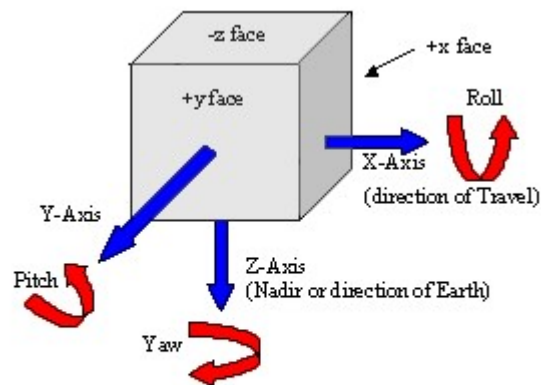
Attitude Estimation by using 3-Axis Rate-Gyro

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Background:

- 'dt' should be very small in order to achieve higher numerical accuracy (Not related to sampling rate).
- Rotation direction sign is derived by the right hand rule.
- The new State-Matrix is solved by multiplying the State-Matrix by the Delta-Matrix ($R_{new} = R \cdot W$)

Roll - ϕ - Phi - Gx
Pitch - θ - Theta - Gy
Yaw - ψ - psi - Gz



Spacecraft Body-Axis Definition

X-Axis - Direction of Travel
Y-Axis - Resultant Direction
Z-Axis - Direction of Earth

Init Conditions:

$$\omega_x := 0 \quad \omega_y := \frac{\pi}{2} \quad \omega_z := 0$$

$$\phi := 0 \quad \theta := 0 \quad \psi := 0$$

$$\underline{\underline{A}} := (0 \ 0 \ 9.8)$$

Delta-Matrix (Build from the Gyros measurements):

$$M_{\omega}(dt) := \begin{pmatrix} 1 & -\omega_z \cdot dt & \omega_y \cdot dt \\ \omega_z \cdot dt & 1 & -\omega_x \cdot dt \\ -\omega_y \cdot dt & \omega_x \cdot dt & 1 \end{pmatrix}$$

State-Matrix - First the DCM matrix is built from init conditions (ψ, θ, ϕ):

$$DCM_R := \begin{pmatrix} \cos(\theta) \cdot \cos(\psi) & \sin(\phi) \cdot \sin(\theta) \cdot \cos(\psi) - \cos(\phi) \cdot \sin(\psi) & \cos(\phi) \cdot \sin(\theta) \cdot \cos(\psi) + \sin(\phi) \cdot \sin(\psi) \\ \cos(\theta) \cdot \sin(\psi) & \sin(\phi) \cdot \sin(\theta) \cdot \sin(\psi) + \cos(\phi) \cdot \cos(\psi) & \cos(\phi) \cdot \sin(\theta) \cdot \sin(\psi) - \sin(\phi) \cdot \cos(\psi) \\ -\sin(\theta) & \sin(\phi) \cdot \cos(\theta) & \cos(\phi) \cdot \cos(\theta) \end{pmatrix}$$

$$DCM_R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{Init value of State-Matrix}$$

Integration (of angular velocities):

$$ROT(R, W, N) := \begin{cases} \text{for } i \in 0..N \\ \quad R \leftarrow R \cdot W \\ \text{return } R \end{cases}$$

R is the State-Matrix.
W is the Delta-Matrix.
N is the number of iterations.

Result:

$$A \cdot ROT(DCM_R, M_{\omega}(0.1), 10) = (-11.091 \ 0 \ -1.598)$$

$$A \cdot ROT(DCM_R, M_{\omega}(0.01), 100) = (-9.922 \ 0 \ -0.155)$$

$$A \cdot ROT(DCM_R, M_{\omega}(0.001), 1000) = (-9.812 \ 0 \ -0.015)$$

$$A \cdot ROT(DCM_R, M_{\omega}(0.0001), 10000) = (-9.801 \ 0 \ -1.539 \times 10^{-3})$$

Notice that as 'dt' becomes smaller the accuracy increases, even with constant omega values.
This isn't a bandwidth issue. Is it a numerical approximation problem.

